

CONTRACT NOTE REWORK

Training & Data Sources

Estimate 3 - Technical Walkthrough

~16.9 developer days (3a: 8.0 + 3b: 8.9)

Overview

Estimate 3 has two independent parts that can be prioritised separately. Part 3a is training and enablement, the documentation and materials that let the BRYT team run the new system themselves. Part 3b is data source extensibility, a software capability that lets business users enrich contract notes with data from new sources without developer involvement.

This document gives a brief summary of 3a, then focuses on the technical detail of 3b, which is the build effort.

3a. Training & Enablement (~8 days)

This part is a documentation and enablement effort rather than software. The goal is to get a small team of business users (roughly 2-5 people) productive on the template management system from Estimates 1 and 2 without ongoing developer support.

We intend to deliver a quick-start guide (zero to first template in under 30 minutes), a set of task-based how-to guides (create a template, update T&Cs, configure a rule, attach a data source), a data field reference cataloguing every field a designer can use, a one-page rules engine cheat sheet, template design-pattern guidance, and a troubleshooting FAQ. Optional screen recordings can follow once the UI is live.

These assets are best produced against the real UI, so drafts are written from the wireframes during the build and finalised with real screenshots afterwards, closing with a live training session and handover.

Why 3a comes after the build

Training materials lean on the actual screens. We draft them early from the mockups, then capture real screenshots and record walkthroughs once Estimates 1 and 3b are implemented, so the guides match what users actually see.

3b. Data Source Extensibility

Contract notes today can only show data that arrives in the contract payload. If the business wants to include something new, a credit score, a loyalty tier, a bespoke figure from another system, that has historically meant a developer change to the pipeline.

Estimate 3b removes that dependency. Business users manage data sources in SageMaker Unified Studio; once a source is subscribed to the contract note project, it becomes automatically discoverable in the Admin Portal. Users attach it to a template, use its fields in the section designer, and at render time the pipeline fetches the matching data and merges it in, all without a code change or redeployment.

The join key throughout is the BrytNumber (the customer reference already present in the contract payload). Every data source must expose a `bryt_number` column so it can be matched to the customer being rendered.

How it works

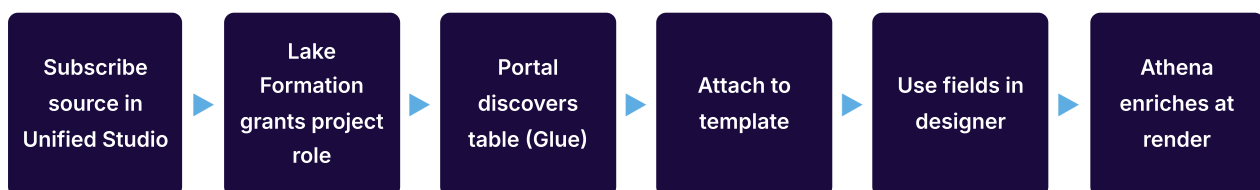
The capability spans three touch points: discovery, design, and render.

Discovery. When a user subscribes a data source in Unified Studio, Lake Formation grants land on the project role. The Admin Portal, assuming that same role, lists the newly available tables from the Glue Data Catalog, showing only those with a `bryt_number` column so they can actually be joined.

Design. A user attaches a data source to a template, and its columns then appear as fields in the section editor, grouped and namespaced by source, alongside the core contract fields. Shared sections automatically track which data sources they depend on.

Render. When the pipeline renders a template, it looks up the attached data sources, queries each one via Athena using the BrytNumber, and merges the results into the contract data before the sections are rendered.

From subscription to rendered field



A newly subscribed data source is usable end-to-end with no code change or redeployment.

The screen

Estimate 3b extends the existing Template Edit screen from Estimate 1 rather than adding new screens. A data sources panel is added, and the section editor gains a data source field group.

Template Edit - Data Sources panel

The screenshot displays the 'Edit Template: Pure Certainty HH' interface. It is divided into three main sections:

- Template Information:** Includes fields for 'Template Name' (Pure Certainty HH) and 'Description' (Fixed HH contract for commercial customers).
- Sections (drag to reorder):** A list of five sections: 1. Header (shared), 2. Contract Summary, 3. Pricing Table, 4. Credit Summary (shared), and 5. Standard T&Cs v6.4 (shared). Each section has an 'Edit in Designer' button and a 'Remove' button. At the bottom are buttons for '+ New Section' and '+ Add Shared Section'.
- Data Sources:** This panel shows two attached data sources: 'credit_data' (3 cols) and 'consumption_...' (5 cols). A warning message states: 'Credit Summary' section requires credit_data. Below this is a '+ Attach Data Source' button. An 'Available Fields Preview' section lists fields for both sources: credit_data (credit_score (int), credit_grade (string), credit_limit (double)) and consumption_forecast (forecast_annual_kwh (double), forecast_peak_kw (double), forecast_method (string), forecast_date (date), confidence_pct (int)).

A 'Save Changes' button is located in the top right corner.

The Template Edit screen gains a Data Sources panel showing which sources are attached to the template. From here a user attaches an available source or detaches one they no longer need.

KEY INTERACTIONS

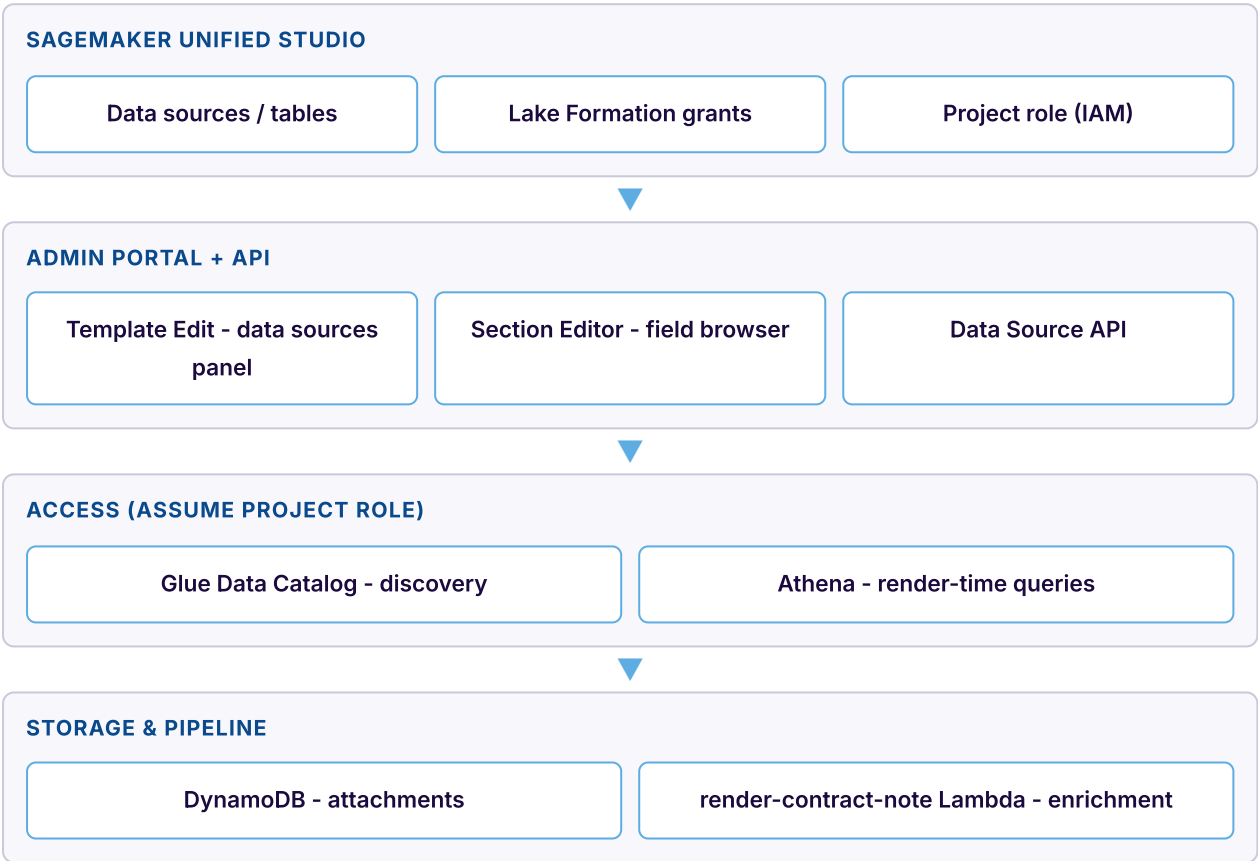
- › Attach Data Source opens a picker of available (unattached) sources
- › Each attached source shows its name and column count
- › Detach warns and lists affected sections if fields are in use
- › Attached source columns become available in the section editor

BEHIND THE SCREEN

- › Attachments are stored as records under the template in the existing DynamoDB table
- › The picker is populated from the Glue catalog via the project role
- › Only tables with a bryt_number column are offered
- › Shared sections separately track their own data source dependencies

System architecture

The Admin Portal and render pipeline both reach data sources by assuming the Unified Studio project role, which already holds the Lake Formation grants. Discovery goes through the Glue Data Catalog; render-time enrichment goes through Athena. Attachment records live in the same DynamoDB table introduced in Estimate 1.



Access to data sources is inherited via the project role, so new subscriptions need no IAM changes.

Key design decisions

The design leans on the existing Unified Studio governance rather than reinventing access control.

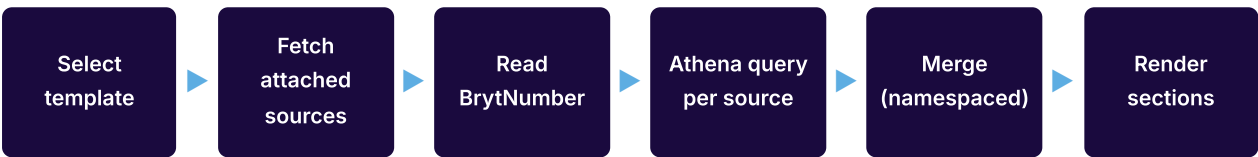
DECISION	CHOICE	WHY
Data source access	Assume the Unified Studio project role	Inherits all Lake Formation grants automatically; no per-table IAM configuration
Catalogue discovery	Glue Data Catalog API	The standard AWS metadata layer; Unified Studio uses Glue underneath

DECISION	CHOICE	WHY
Render-time query	Athena	Serverless SQL over Glue/Iceberg tables; handles varied storage formats
Field namespaces	{source}.{column}	Avoids collisions between data sources and core contract fields; makes provenance clear
Template binding	Explicit attachment per template	Limits render-time queries to what's actually needed; keeps dependencies visible
Join constraint	Table must have a bryt_number column	Enforces join-ability and filters out tables that can't be matched to a customer
Section dependencies	Auto-tracked from field references	No manual config; derived by scanning which data source fields a section uses

Render-time enrichment

Enrichment slots into the existing render pipeline as a single new step between template selection and section rendering. Once a template is chosen, the pipeline gathers its attached data sources, reads the BrytNumber from the contract data, and queries each source in parallel via Athena. The results are merged into the contract data under a per-source namespace, then rendering proceeds exactly as before.

Enrichment step in the pipeline



Queries run in parallel to keep added latency low.

How enrichment handles the edge cases

SITUATION	BEHAVIOUR
Row found for the BrytNumber	Columns merged in under {table}.{column} and available to fields
No row for the BrytNumber	Warning logged; rendering continues with those fields empty
Multiple rows returned	First row used; a warning is logged
Athena query fails or times out	Error logged to the error bucket; rendering halts (no partial PDF)

What we store

No new table is needed. Two record types are added to the existing ContractNoteTemplates table.

RECORD	HOLDS
Template data source	The database + table attached to a template, display name, and who attached it / when

RECORD	HOLDS
Shared section dependency	The data sources a shared section requires, derived from the fields it uses
Field reference (in schema)	Data source fields are stored namespaced, e.g. <code>credit_data.credit_score</code>

Delivery breakdown (3b)

Estimate 3b is ~6.4 required days plus optional testing, ~8.9 days in total. Part 3a adds ~8 days of training/enablement effort.

AREA	SCOPE
Infrastructure & IAM	Project role trust policy for Lambda assumption, Athena workgroup + results bucket, DynamoDB record types
Glue discovery client	Assume project role, list tables, filter to bryt_number tables, fetch column detail
Data Source API	List available, get columns, attach, detach (with field-in-use check), list attached
Render enrichment	Athena query executor, enrichment orchestrator, integration into the render pipeline
Template Edit panel	Data sources panel, attach/detach with warnings, data source picker dialog
Section editor fields	Data source field group, dependency tracking, missing-dependency prompt, dependency display
Integration	CDK wiring for routes, role assumption, Athena, and end-to-end validation

A note on cost

Athena is billed by data scanned. For large data sources it is worth partitioning or pruning on bryt_number so each render-time lookup scans as little as possible. Queries also run in parallel across sources to keep the added render latency down.

Testing note

3b is designed around 10 correctness properties (bryt_number-only discovery, attachment round-trips, namespaced enrichment, graceful handling of missing rows, and fail-safe on query errors). Property-based and integration tests are optional in the plan and can be deferred for a faster MVP, but are recommended given enrichment feeds live customer documents.